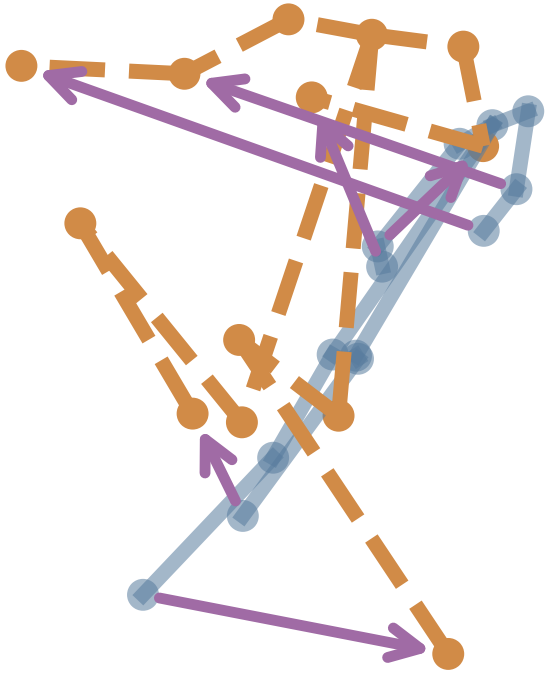Output residual correction